

Week 1 – TA Session

When Can We Trust an Estimate?

Course: Machine Learning for Signal Processing

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Part I

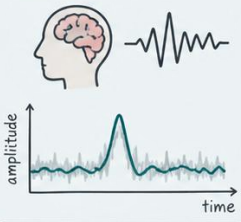
From Noisy Observations to Statistical Estimation

One Estimation Problem, Many Applications

A measurement system produces a characteristic signal whose shape is known from calibration or previous measurements, but its amplitude is unknown and our observations are corrupted by noise..

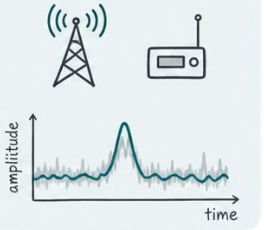
Biomedical Engineering

Estimating the amplitude of an evoked-response waveform.



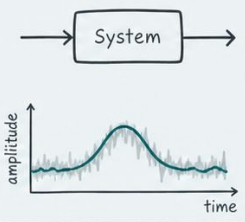
Telecommunications

Estimating the amplitude of a received reference signal.



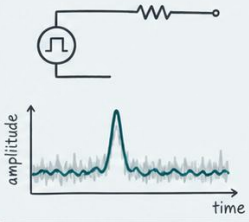
Control Systems

Estimating the magnitude of a known system response.



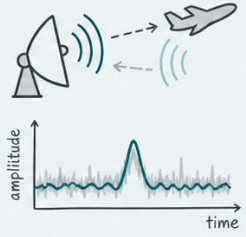
Electronics

Measuring the amplitude of a transient pulse.



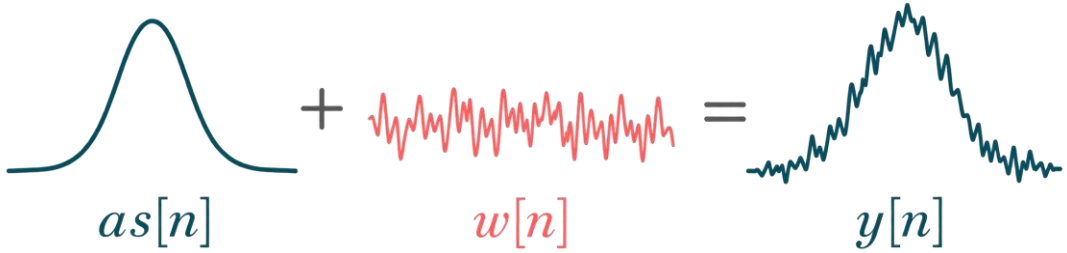
Radar

Estimating the strength of a received echo.



How can we estimate the amplitude of a known waveform from noisy observations?

Can We Recover a Signal from Noisy Observations?



Model

$$y[n] = as[n] + w[n]$$

- $s[n]$: Known signal
- a: Unknown amplitude
- $w[n]$: Additive Gaussian noise
- $y[n]$: Observed signal

Goal

Estimate a from $y[n]$, and determine how reliable that estimate is.

Q1:

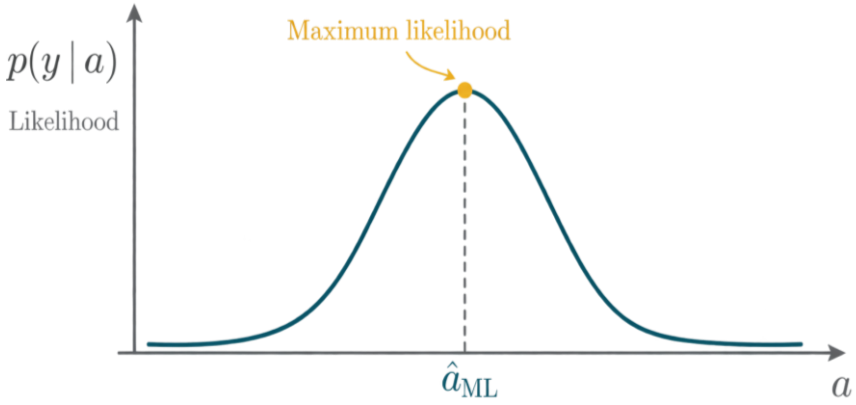
If we observe the same signal twice, with different noise realizations, should we obtain the same amplitude estimate?

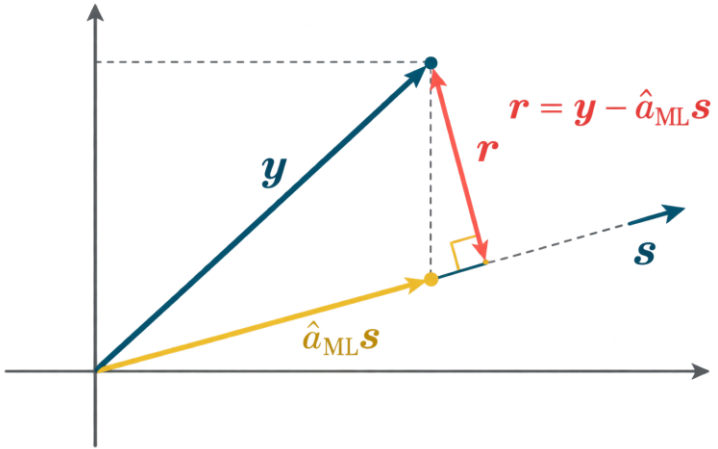
Maximum Likelihood as a Geometric Projection

- Maximum likelihood estimation:

$$\widehat{a_{\mathrm{ML}}} = \underset{a}{\operatorname{argmax}}\, p(y|a)$$

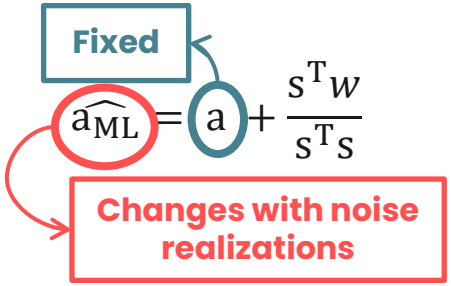
- *ML asks which parameter value best explains the observation.*





EXPERIMENT 1

How Reliable Is Our Estimate?

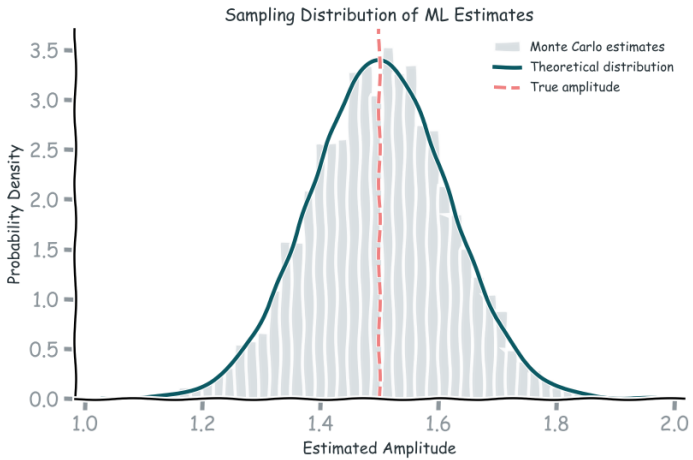


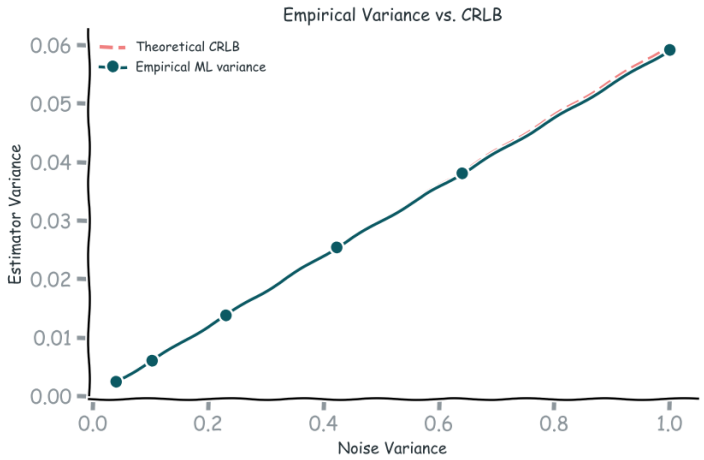
The Cramér–Rao lower bound provides a theoretical variance benchmark for unbiased estimators under appropriate assumptions.

Q2:

If we double the measurement noise standard deviation, how much will the variance of our amplitude estimates increase?

A. No change
B. Twice as large
C. Four times as large
D. Cannot be determined


Sampling Distribution of ML Estimates
Monte Carlo estimates
Theoretical distribution
True amplitude
Probability Density
3.5
3.0
2.5
2.0
1.5
1.0
0.5
0.0
1.0
1.2
1.4
1.6
1.8
2.0
Estimated Amplitude

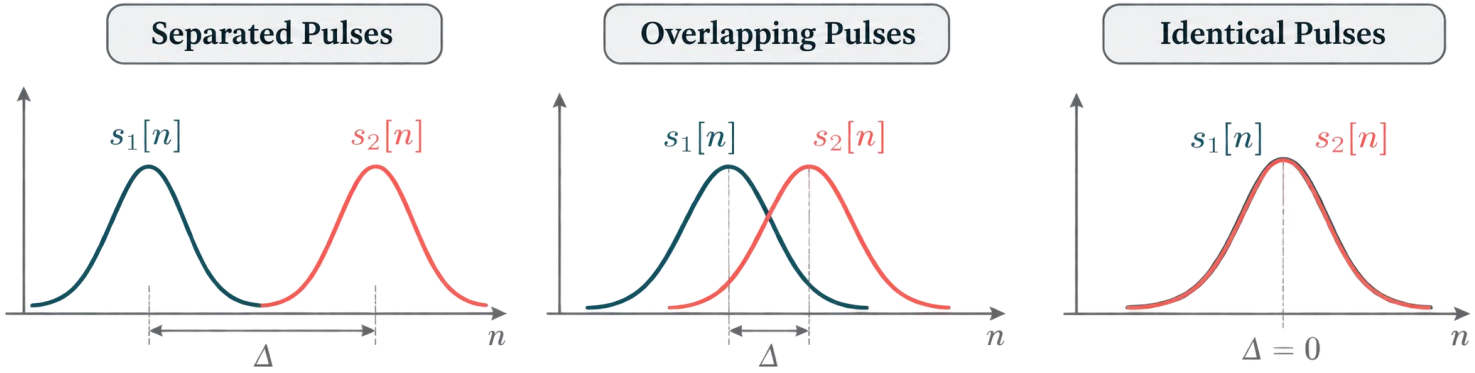

Empirical Variance vs. CRLB
Theoretical CRLB
Empirical ML variance
Estimator Variance
0.06
0.05
0.04
0.03
0.02
0.01
0.00
0.0
0.2
0.4
0.6
0.8
1.0
Noise Variance

EXPERIMENT 2

Part II

When Maximum Likelihood
Becomes Unreliable

When Estimation Becomes Difficult



Model

$$y[n] = a_1 s_1[n] + a_2 s_2[n] + w[n]$$

Goal

Estimate the two unknown amplitudes a_1 & a_2.

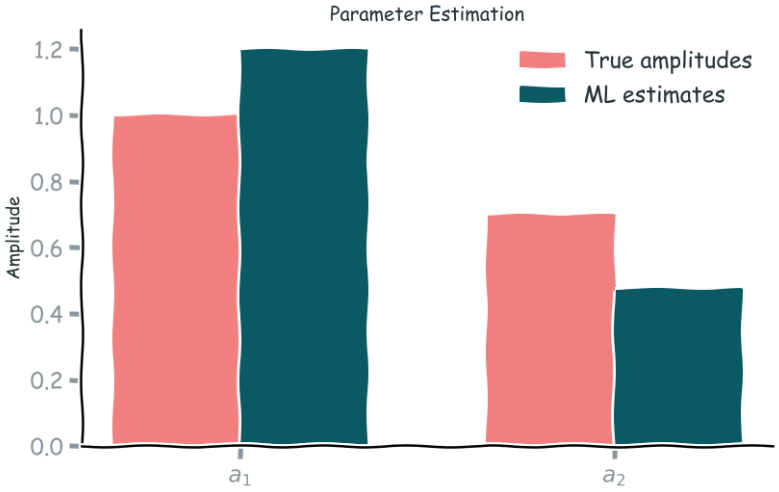
Q3:

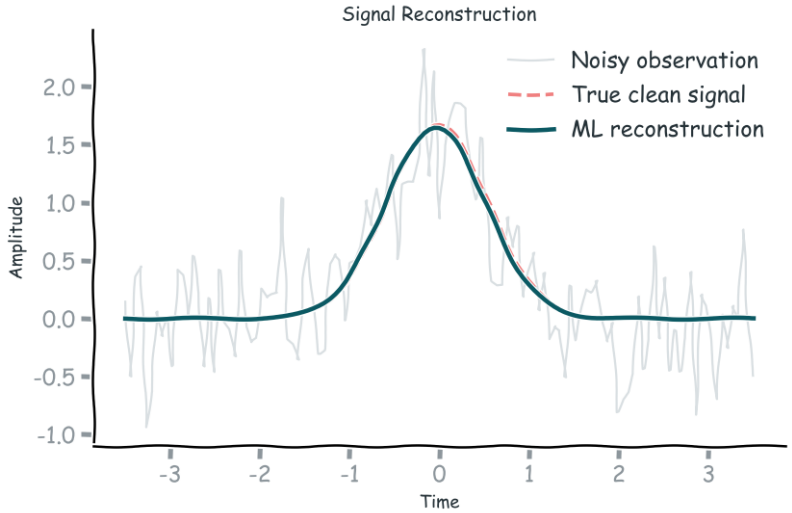
If the two pulses become identical, can we still estimate their individual amplitudes? Can we still reconstruct their combined signal?

EXPERIMENT 3

A Good Reconstruction Is Not Always a Good Estimate

When signal templates are highly similar, inaccurate amplitude estimates can still produce accurate signal reconstructions.





Part III

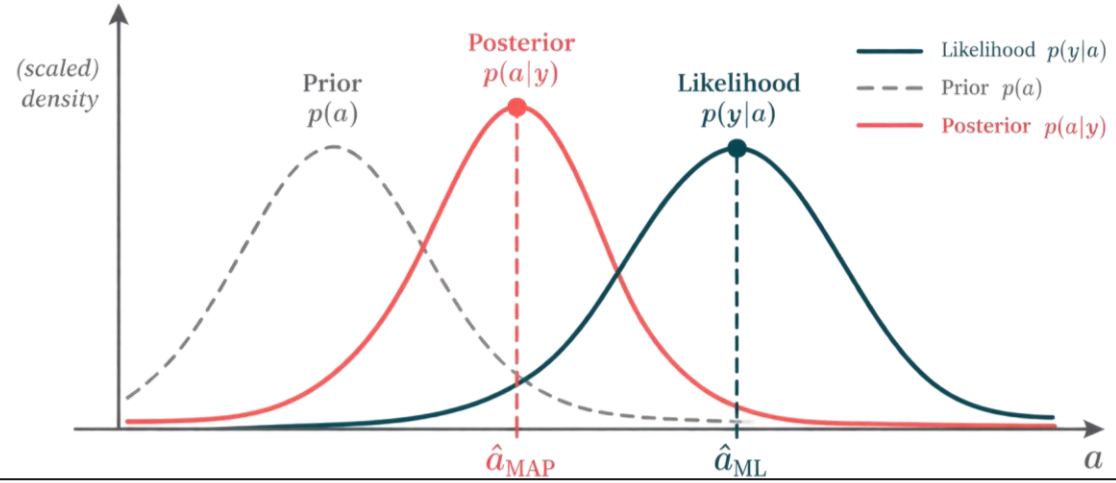
Bayesian Estimation

Maximum a Posteriori Estimation (MAP)

- Observed Data + Prior Knowledge → Posterior Distribution

$$\widehat{a_{MAP}} = \underset{a}{\operatorname{argmax}}\, p(a|y)$$

- *MAP asks which parameter value is most plausible after incorporating both the observations and prior information*



Q4:

If our measurements are very noisy, but we have strong prior information, Which will have more influence on the MAP estimate?

EXPERIMENT 4

Part IV

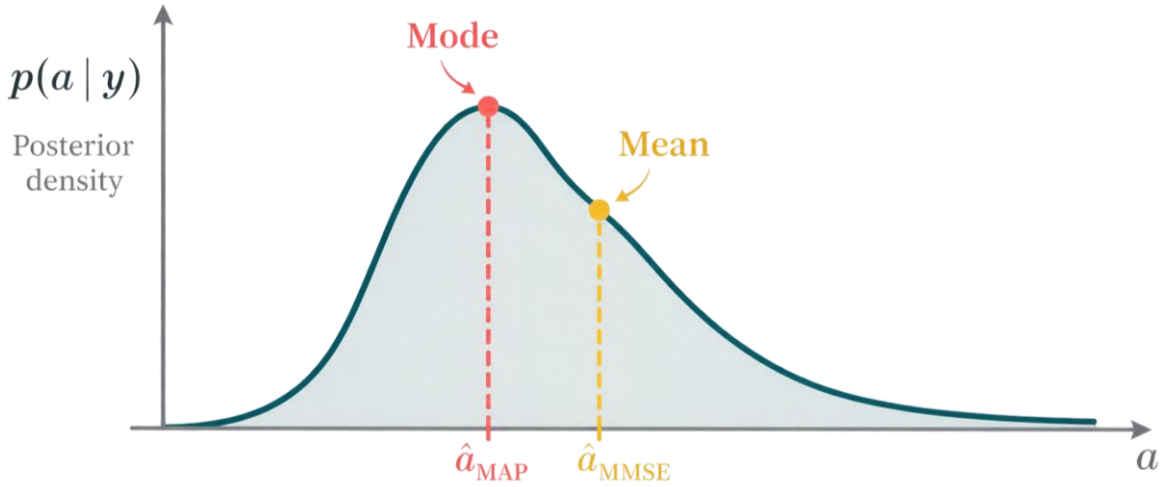
MMSE

Minimum Mean Squared Error (MMSE)

- The MMSE estimator chooses the posterior mean.

$$\widehat{a_{MMSE}} = \mathbb{E}[a|y]$$

- *MMSE asks which estimate minimizes the posterior expected squared error.*



EXPERIMENT 5

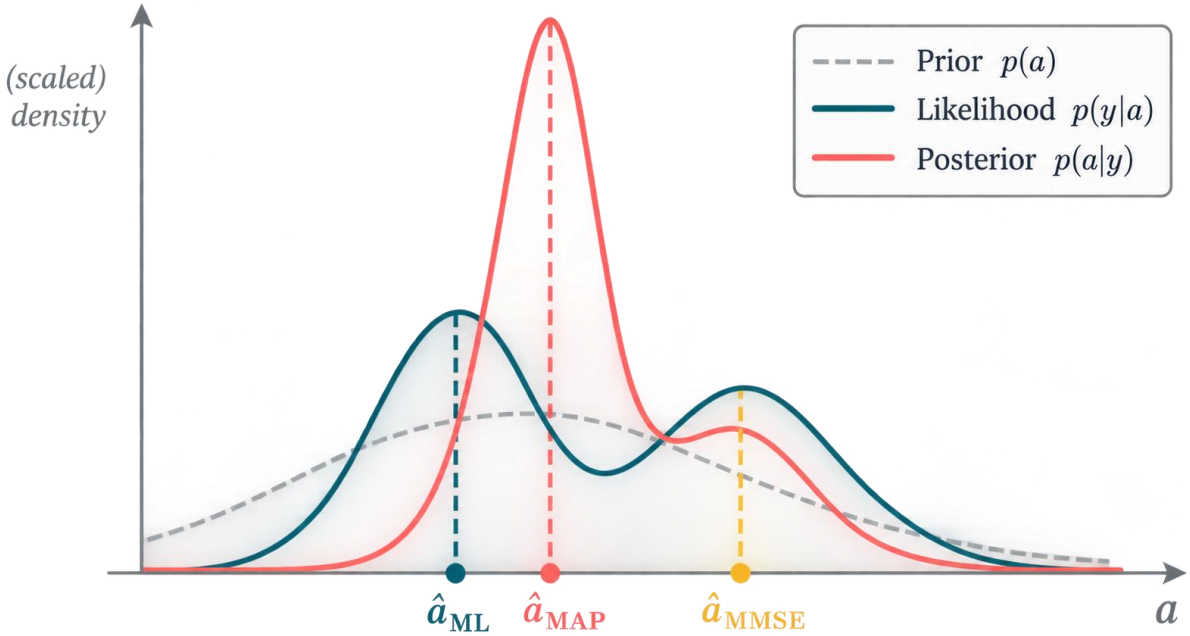
Conclusion

Final Comparison

Final Comparison

	ML	MAP	MMSE
Main question	What best explains the observed data?	What is most plausible after observing the data?	What minimizes posterior expected squared error?
Uses a prior	No	Yes	Yes
Based on	Likelihood	Posterior mode	Posterior mean
Main strength	Does not require a prior	Incorporates prior information	Optimal under posterior squared-error loss
Main limitation	Can be unstable with insufficient information	Sensitive to prior assumptions	Requires computing a posterior expectation

ML vs. MAP vs. MMSE



THANKS!

Do you have any questions?

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